

CS3300 A Introduction to Software Engineering

Lecture 04: SDLC; Life Cycle Models

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Traditional Software Development Phases



Requirements

Engineering



Design



Implementation



Verification &
Validation



Maintenance

Software Development Phases: Semester Assignments



Requirements
Engineering

Project 1 Planning; Project 1 RE



Design

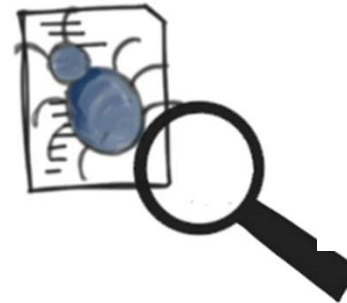
Project 1 Design

AI integration



Implementation

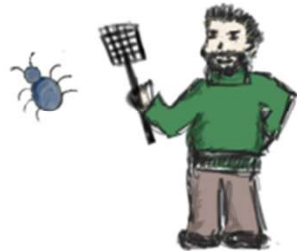
Project 1 & 2 code, report, ppt



Verification & Validation

AI integration

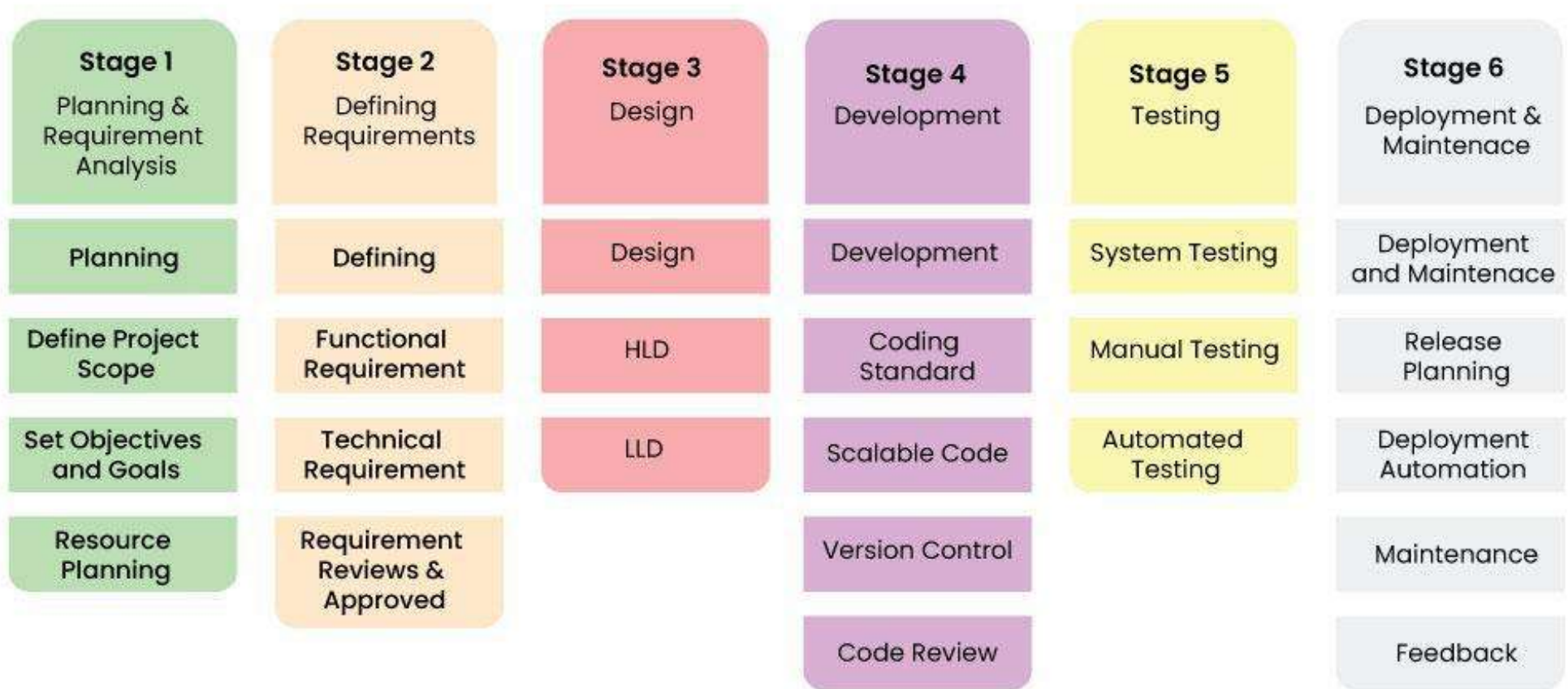
Project 2 Test



Maintenance



Software Development Life Cycle



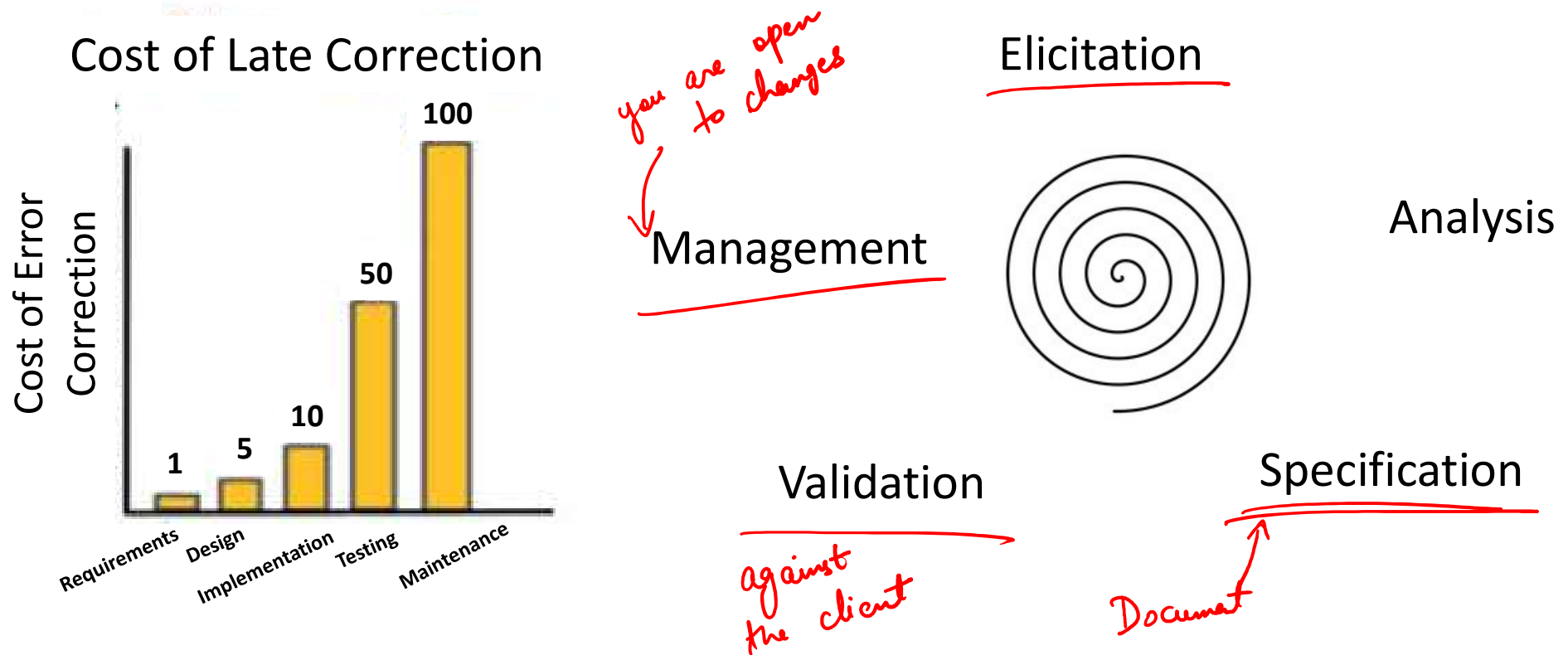
6 Stages of Software Development Life Cycle



Requirements Engineering



RE is the process of establishing the needs of stakeholders that are to be solved by software



RE Example: Task Management Software

Elicitation:

Conduct interviews with potential users (project managers, team members) to gather what features they need in a task management tool. For example, task assignment, deadlines, notifications, and progress tracking might be common requirements.



Analysis:

Analyze the feedback to determine essential features versus nice-to-have features. This might involve grouping similar requirements and identifying conflicts or unrealistic expectations.



Specification:

Create a detailed **SRS** document that clearly lists all functional and non-functional requirements, such as user roles, security levels, and user interface preferences.

Software Requirements Specification



Validation:

Review the SRS with stakeholders to ensure the requirements are complete, consistent, realistic, and verifiable. Make Modifications based on feedback to better align with user expectations and technical feasibility.



Management:

Establish a process to accommodate changes in requirements throughout the project lifecycle, ensuring traceability and controlled integration of changes.

Design



SRS (Software Requirements Specification) is a reference for software designers to come up with the best design for the software.

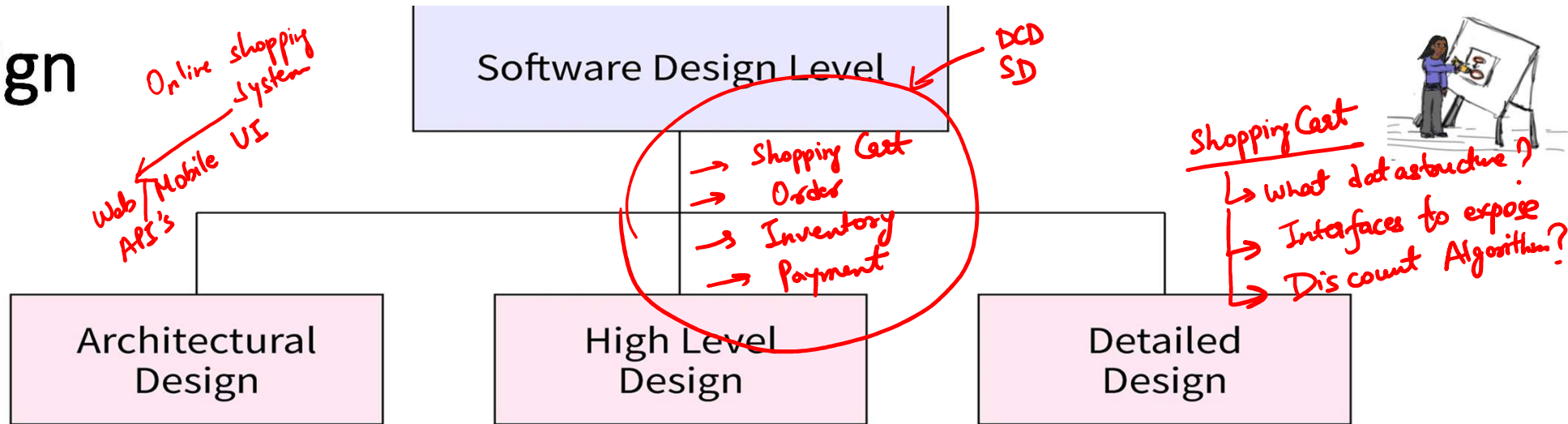


Multiple designs for the product architecture are present in the Design Document Specification (DDS).



This DDS is assessed by market analysts and stakeholders. After evaluating all the possible factors, the most practical and logical design is chosen for development.

Design



- Defines the overall structure of the system.
 - Identifies the major components or subsystems, their responsibilities, and how they are connected.
 - **Focus:** What are the major pieces of the system?
- Breaks major components into smaller modules/classes.
 - Defines their responsibilities, relationships, and interactions with one another.
 - **Focus:** What is inside each major component, and how do the pieces interact?
- Defines how individual modules/classes will be implemented.
 - Specifies methods, interfaces, data structures, and algorithms needed to implement each component.
 - **Focus:** How exactly will each piece work?

RE and Design Example: Task Management Software

Architectural Design:

Define the overall structure of the system. For this task management software, you might decide on a web-based architecture with client-server model where the **server handles logic and database** interactions, and the **client provides interactive user interfaces**.



High-Level Design:

Break down the architecture into major components or modules such as **User Management, Task Management, Notification System, and Database**. Define the relationships and data flow between these modules.



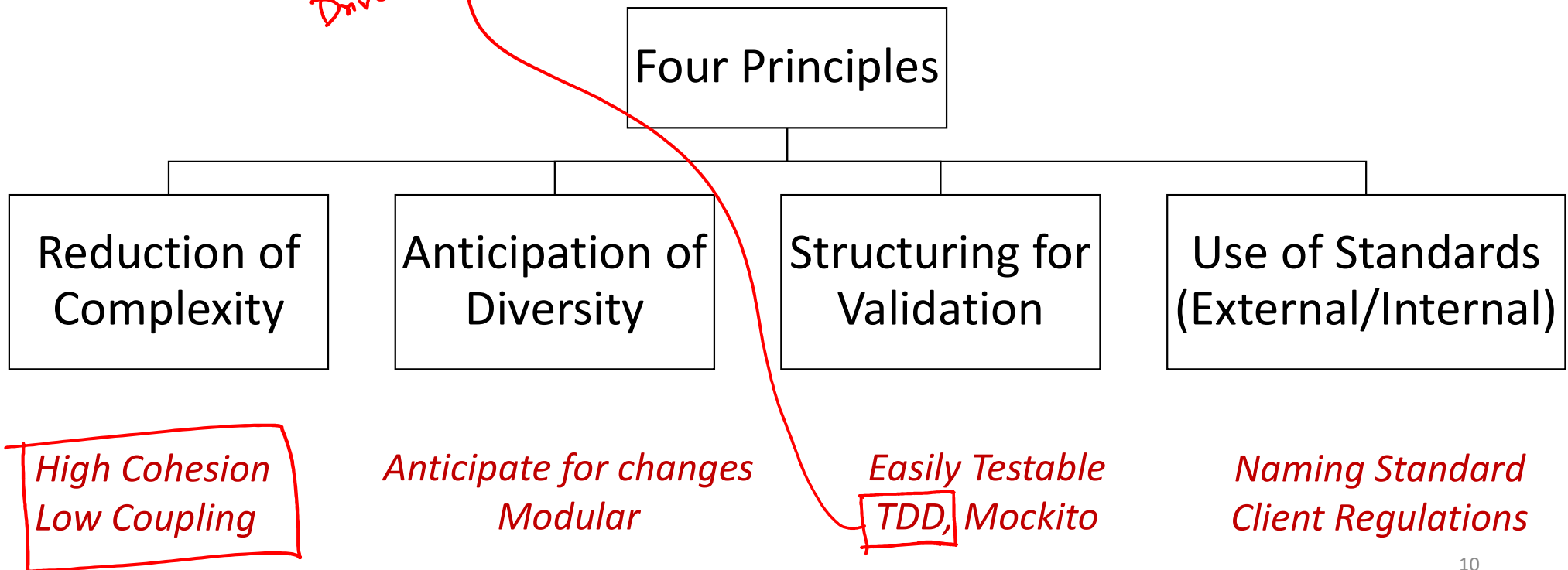
Detailed Design: Focus on the specifics of each module. For instance, the **Task Management module** might involve detailed designs of the database **schema for tasks, classes, and methods to handle task creation, updates, and queries**. Interfaces for each module should also be defined to ensure they can interact seamlessly.

Implementation



Phase where we take care of realizing the design of the system and create a natural softer system

Test Driven Development



Verification & Validation



Phase that aims to check that software system meets its specifications and fulfils its intended purpose

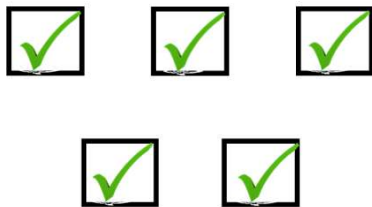
Verification: did we build the system right?

Validation: did we build the right system?

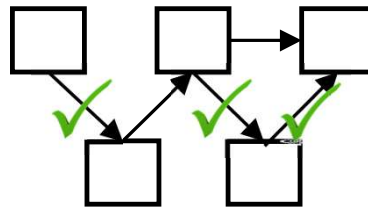
development system

against the client's understanding

Unit



Integration



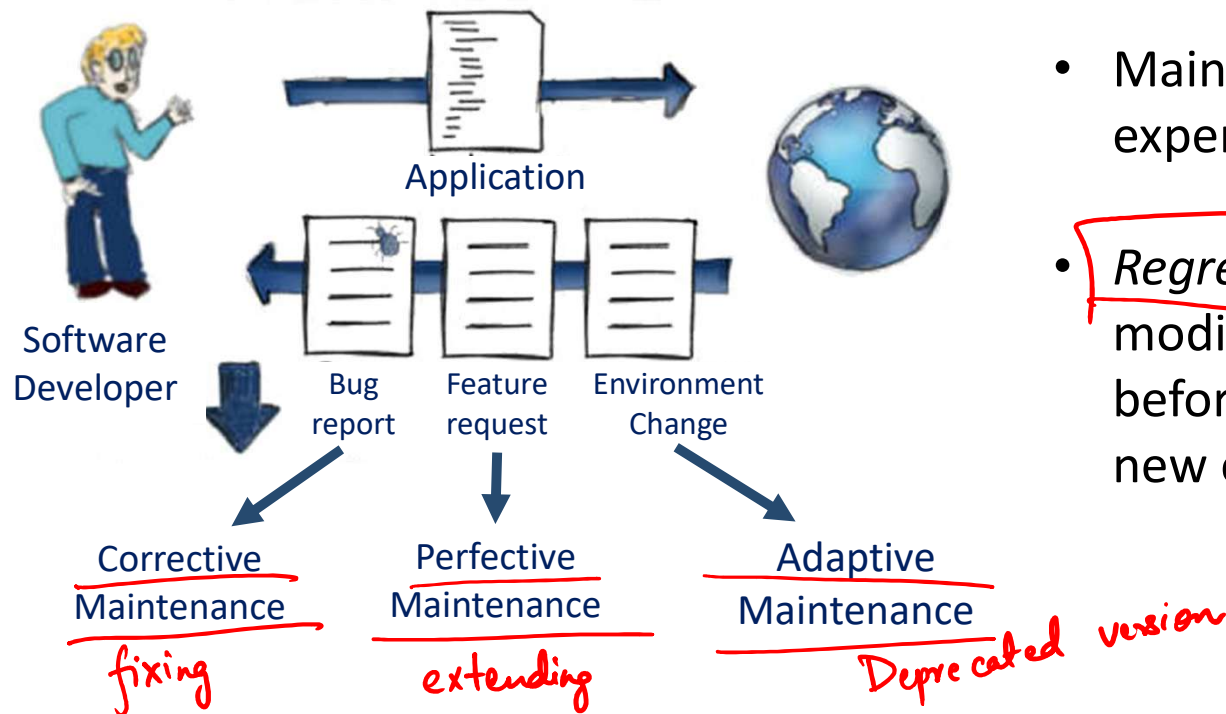
System



Maintenance

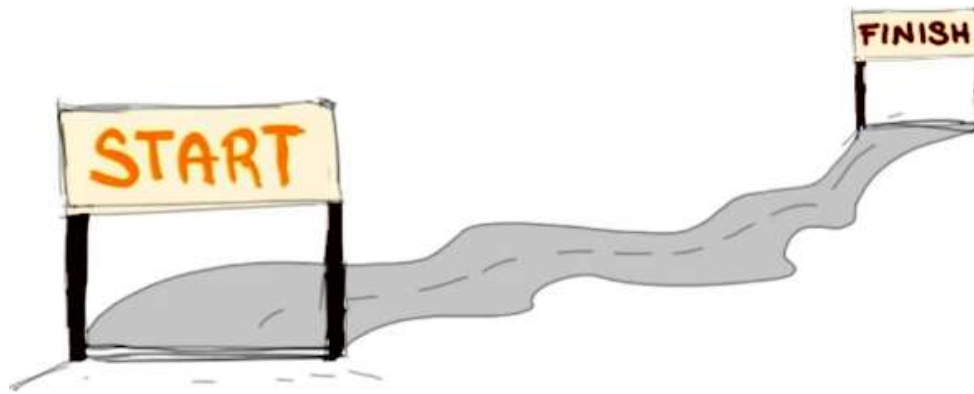


Once Software released to final users and in operation, many things can happen: environment change -new libraries, new systems, additional functionality requests, bug reports



- Maintenance is a fundamental and expensive phase
- Regression testing – retesting a modified version of software before release, no introduction of new errors

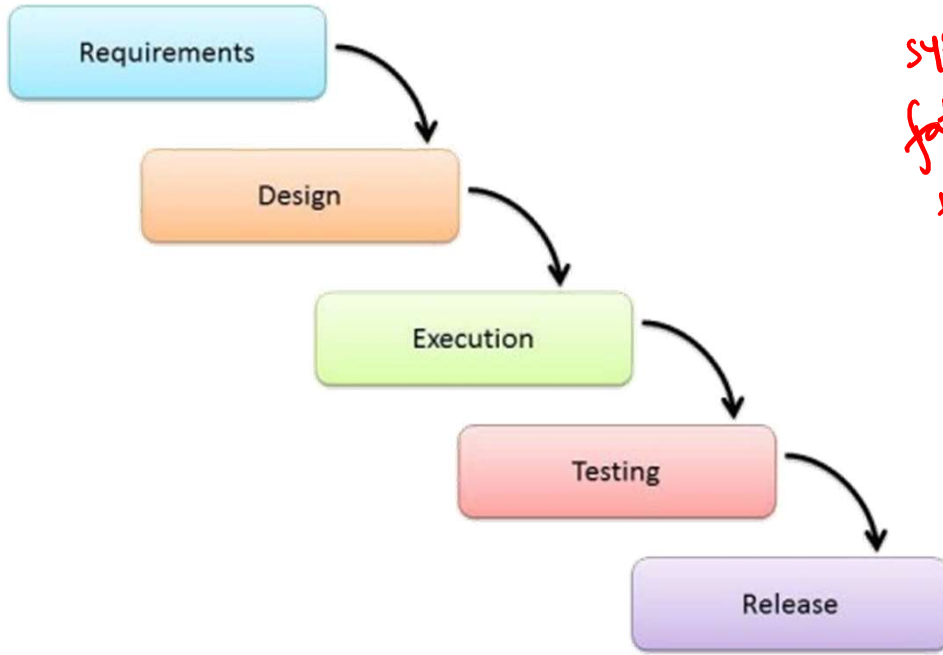
Software Process Model/ Life Cycle Model



Functions:

- Order of activities
- Transition Criteria between Activities
- What should we do next and for how long?

Waterfall Method



*Human life is at stake
system failure
fatality is more important
than money & time.*

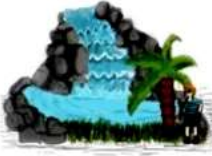


Early Error Detection



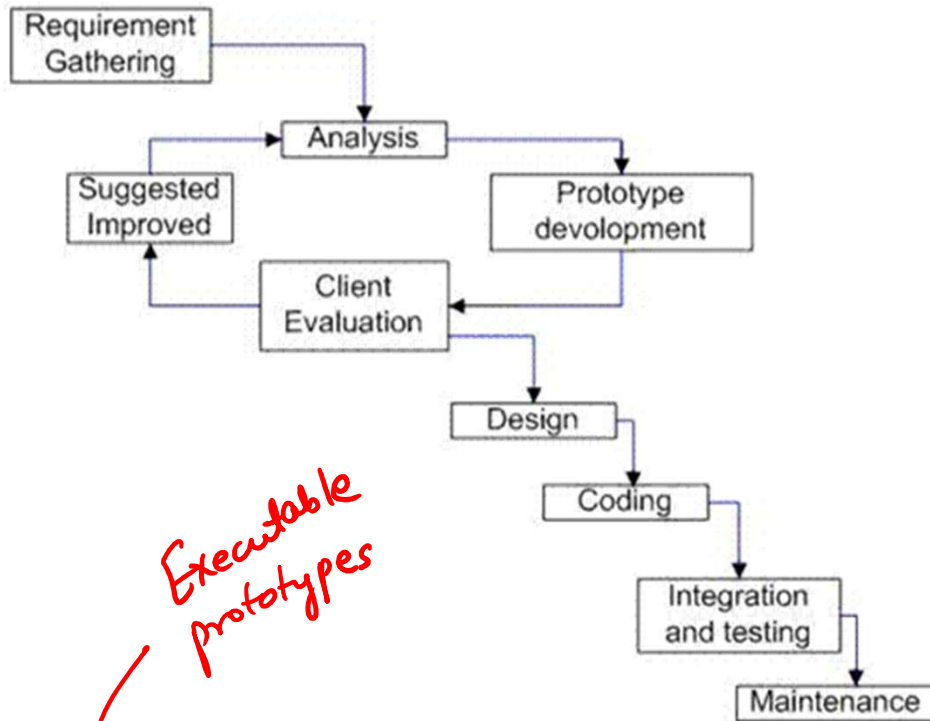
No Flexibility

*Rocket
Aircraft
Medical*



- Project progresses in an orderly sequence of steps
- Pure Waterfall model performs well for software products with a stable product definition- well known domain, technologies involved, Request for Proposals (RFP)
- Waterfall method finds errors in early local stages
- Not flexible- not for projects where requirements change, developers not domain experts, or technology used are new and evolving

Evolutionary Prototyping



Immediate feedback
Helps Requirements understanding



Difficult to Plan
Can deteriorate to code-and-fix

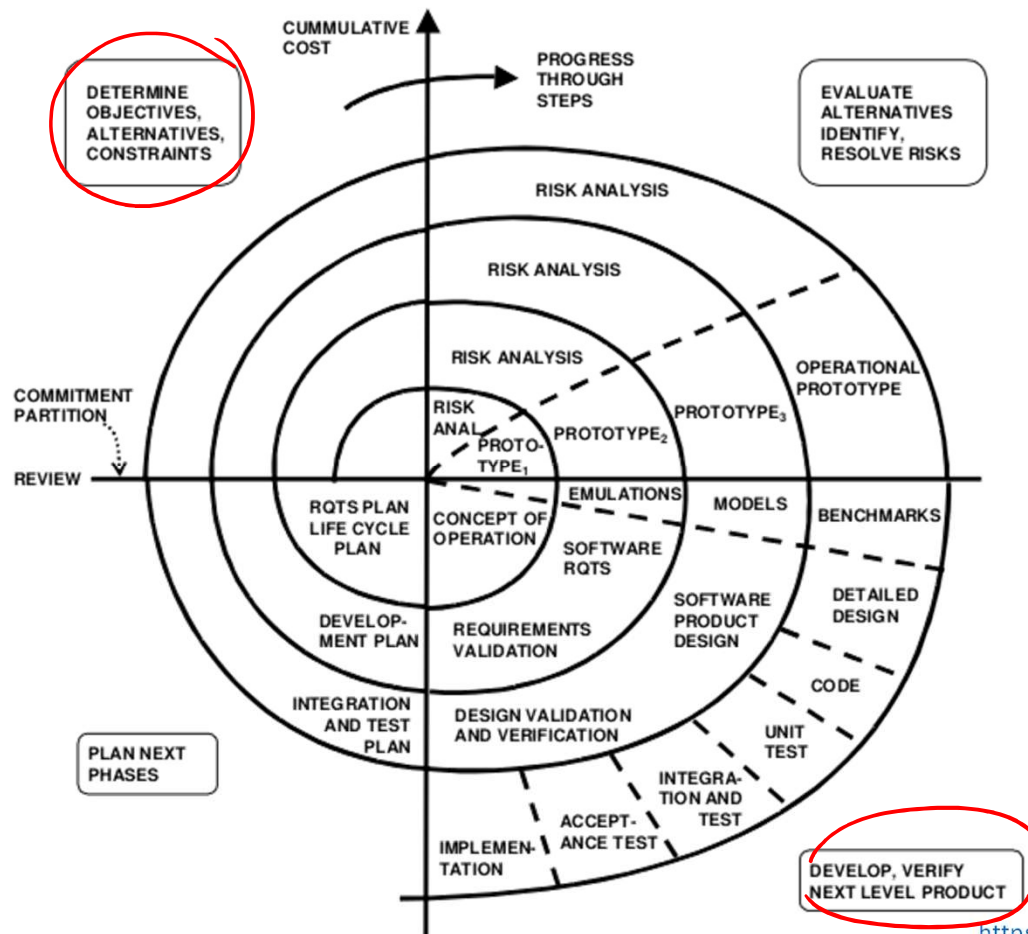
Executable prototypes

- Prototypes that evolve into the final system through an iterative incorporation of user feedback.
- Ideal when not all requirements are well-understood. System keeps evolving based on customer feedback

Spiral Method



Incremental risk-oriented lifecycle model with 4 main phases



Risk Reduction

Functionality can be added
Software produced early, Early feedback



Specific Expertise

Highly dependent on risk analysis
Complex, Costly

Spiral Method



Example: Software for a self-driving car (Deal with biggest risks first)

Cycle 1 — Can we reliably detect lanes?

Objective: Lane detection

Major risk: Cameras may perform poorly in rain/night conditions

Risk reduction: Experiment with different cameras/algorithms and build a prototype

Develop/test: Implement basic lane detection and test it

Next cycle: Decide whether the approach is viable

Cycle 2 — Can we detect pedestrians safely?

Objective: Add pedestrian detection

Major risk: Missing a pedestrian is a critical safety failure

Risk reduction: Test models against difficult scenarios before committing to one

Develop/test: Integrate the selected approach into the existing system

Next cycle: Evaluate results and determine what comes next

Cycle 3 — Can we safely brake?

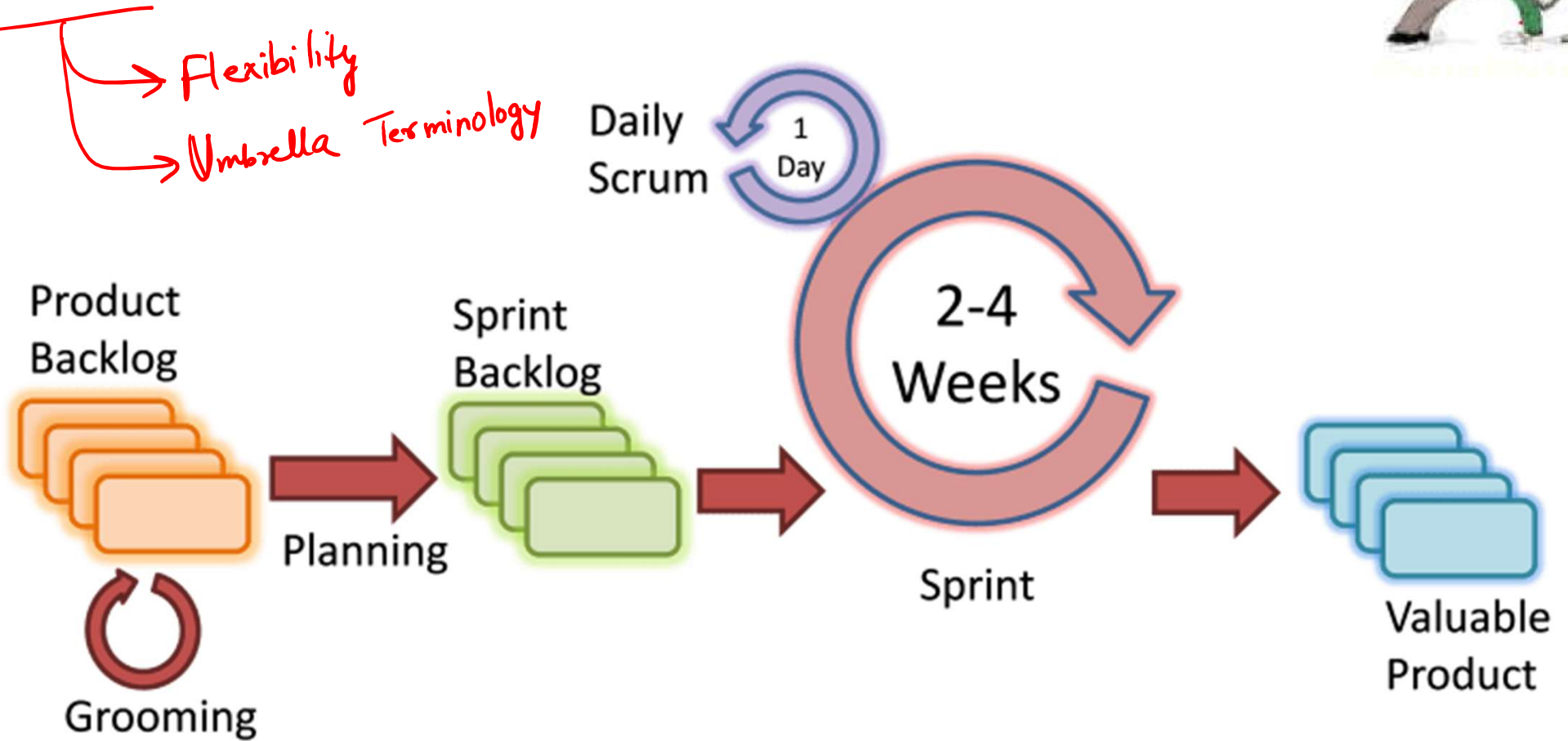
Objective: Automatic emergency braking

Major risk: Braking too late—or braking incorrectly

Risk reduction: Simulation, modeling, controlled testing

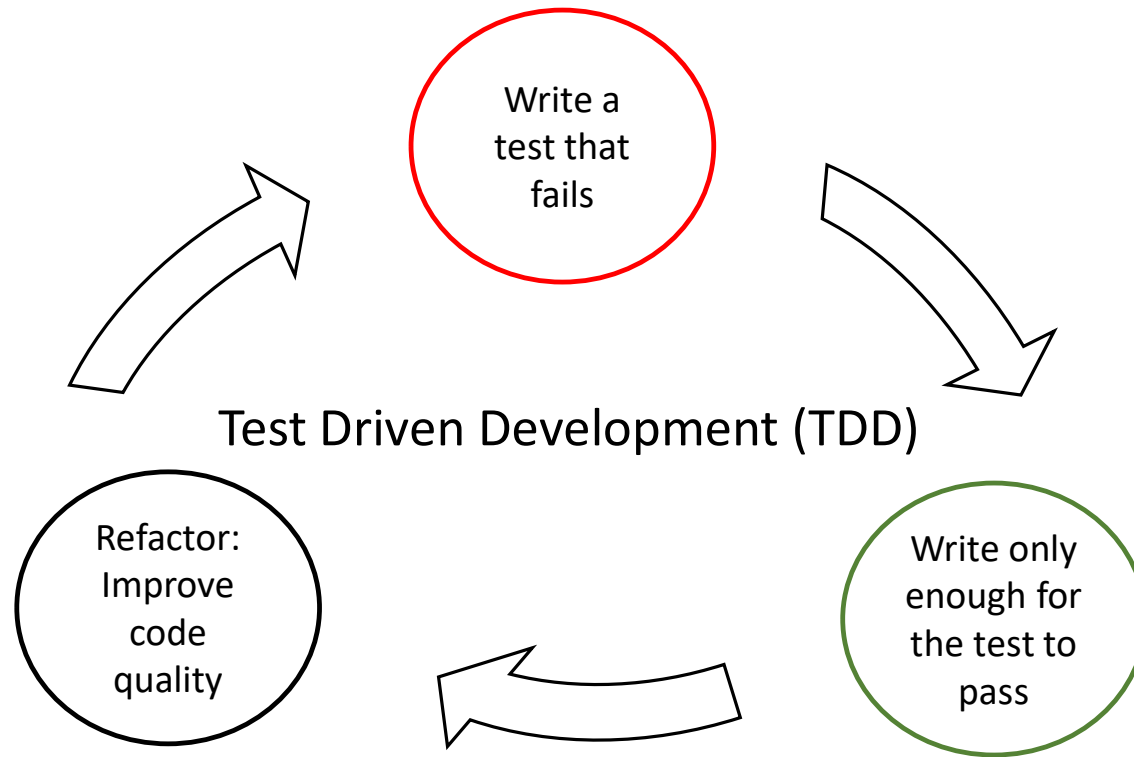
Develop/test: Add braking functionality and integrate it with pedestrian detection

Agile - Scrum



Highly iterative and incremental development process

Agile - XP



- Highly iterative and incremental development process
- Focus on Continuous Integration

Other Agile Methodologies

Kanban: Simplest in IT World;
May Pose time related problems



Some industry-based examples

Waterfall

Military And Aircraft Programs Where Requirements Are Declared Early On And Remain Considerably Constant



Evolutionary Prototyping

- **Company:** Broderbund Software.
- **Project:** The creation of the original "**Prince of Persia**" video game. The initial version of the game was created and then improved upon based on feedback and playtesting.



Some industry-based examples

Spiral

- [NASA's space shuttle program](#) in the 1970s



Agile

- **Apple, IBM, Microsoft, and Procter & Gamble**
- **Cisco:** defects were reduced by 40% when compared to waterfall
- **Barclays:** 300% increase in throughput
- **Panera Bread:** 25% increase in company sales
- **PlayStation Network:** Saved the company \$30 million a year

Companies generally report real benefits from Agile, but they also report significant difficulty implementing and scaling it - especially in larger organizations.

Choosing the right Software Process Model



Requirements
Understanding



Expected
Lifetime



Risk



Schedule Constraints



Interaction with
Management/Customers



Expertise

As much influence over a project's success as any other major planning decision

Industry Standards: Factors affecting choice of project LCM

Degree of
Project
Complexity

Work/Time
Flexibility

Project Focus/
Client
involvement

Size of
organization

Role
Specialization

Budget

<https://asana.com/resources/project-management-methodologies>

<https://thedigitalprojectmanager.com/projects/pm-methodology/project-management-methodologies-made-simple/>

Industry Standards: Factors affecting choice of project LCM

Factors	Waterfall	Evolutionary Prototyping	Agile Methodologies	Spiral
Unclear User Requirements	Poor	Good	Excellent	Excellent
Unfamiliar Technology	Poor	Excellent	Poor	Excellent
Complex System	Good	Excellent	Poor	Excellent
Reliable System	Good	Poor	Good	Excellent
Short time schedule	Poor	Good	Excellent	Excellent
Strong Project Management	Excellent	Excellent	Excellent	Excellent
Cost Limitation	Poor	Poor	Excellent	Poor
Visibility of stakeholder	Good	Excellent	Excellent	Excellent
Skills Limitation	Good	Poor	Poor	Poor
Documentation	Excellent	Good	Poor	Good
Component Reusability	Excellent	Poor	Poor	Poor

<https://melsatar.blog/2012/03/15/software-development-life-cycle-models-and-methodologies/>

Industry Standards: Most Popular Methods

1		Agile – collaborating to iteratively deliver whatever works	6		eXtreme Programming (XP) – doing development robustly to ensure quality
2		Scrum – enabling a small, cross-functional, self-managing team to deliver fast	7		Waterfall – planning projects fully, then executing through phases
3		Kanban – improving speed and quality of delivery by increasing visibility of work in progress and limiting multi-tasking	8		PRINCE2 – controlled project management that leaves nothing to chance
4		Scrumban – limiting work in progress like Kanban, with a daily stand up like Scrum	9		PMI's PMBOK – applying universal standards to Waterfall project management
5		Lean – streamlining and eliminating waste to deliver more with less			

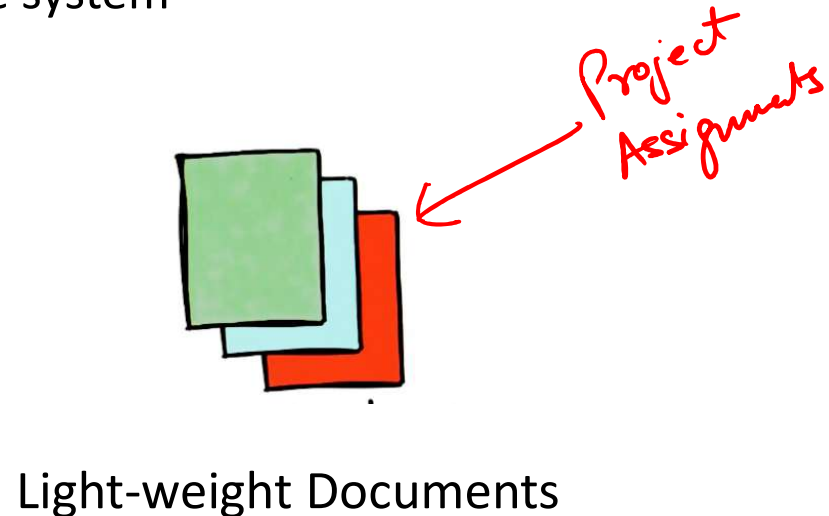
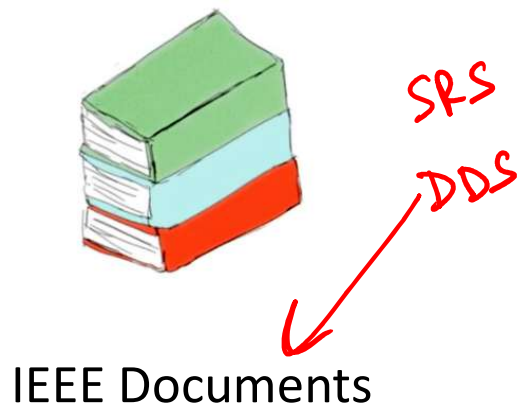
<https://thedigitalprojectmanager.com/projects/pm-methodology/project-management-methodologies-made-simple/>

Lifecycle Documents

Documenting the activities carried out during the different phases of the lifecycle is a very important task.

Can be used for different purposes like:

- Communicate details of the software systems to different stakeholders
- Ensure the correct implementation of the system
- Facilitate maintenance and so on.



Classic Mistakes : People



Heroics



Work Environment



People Management

Classic Mistakes : Process



Schedule Issues



Planning Issues

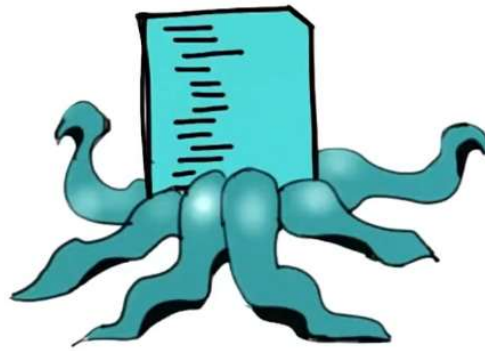


Failure

Classic Mistakes : Product



Gold Plating of
Requirements



Feature Creep

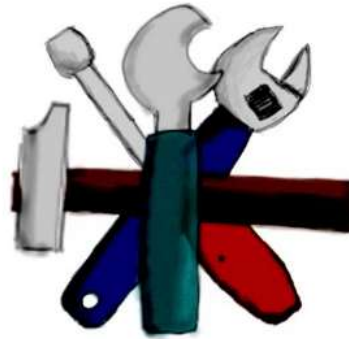


Research ≠ Development

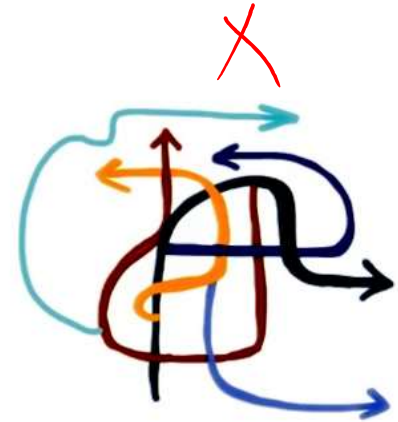
Classic Mistakes : Technology



Silver-Bullet Syndrome



Switching Tools



No version control